Adversarial Bootstrapped Question Representation Learning for Knowledge Tracing

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ABSTRACT

Knowledge tracing (KT), which estimates and traces the degree of learners' mastery of concepts based on students' responses to learning resources, has become an increasingly relevant problem in intelligent education. The accuracy of predictions greatly depends on the quality of question representations. While contrastive learning has been commonly used to generate high-quality representations, the selection of positive and negative samples for knowledge tracing remains a challenge. To address this issue, we propose an adversarial bootstrapped question representation (ABQR) model, which can generate robust and high-quality question representations without requiring negative samples. Specifically, ABQR introduces the bootstrap self-supervised learning framework, which learns question representations from different views of the skill-informed question interaction graph and facilitates question representations between each view to predict one another, thereby circumventing the need for negative sample selection. Moreover, we propose a multi-objective multi-round feature adversarial graph augmentation method to obtain a higher-quality target view, while preserving the structural information of the original graph. ABQR is versatile and can be easily integrated with any base KT model as a plug-in to enhance the quality of question representation. Extensive experiments demonstrate that ABQR significantly improves the performance of the base KT model and outperforms state-of-the-art models. Ablation experiments confirm the effectiveness of each module of ABQR. The code is available at https://github.com/lilstrawberry/ABQR.

CCS CONCEPTS

• **Applied computing** → **Education**; **Distance learning**.

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KEYWORDS

knowledge tracing; contrastive learning; adversarial learning; question representation

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1 INTRODUCTION

With the rapid development of digital multimedia technology, multimedia electronic equipment is becoming increasingly popular, and distance education is becoming a more viable option. As a result, online education formats such as large-scale open online classroom platforms and intelligent tutoring systems are flourishing. Students can learn and practice knowledge using exercises provided by the platform. Due to the increasing number of students studying online, there is a need for a method tracing the state of students' knowledge. By understanding the evolution of a student's knowledge state, the platform can provide personalized learning paths that are tailored to the student's needs. Knowledge tracing (KT) is a fundamental task for an online education platform. It aims to trace a student's mastery of knowledge and predict the probability of the student correctly answering new questions based on their historical performance answering questions[9] (as shown in Fig. 1).

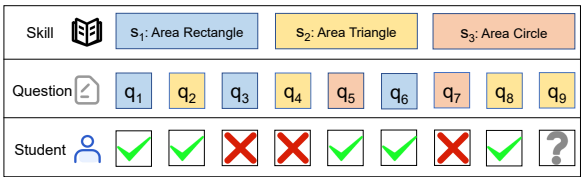

Figure 1: A simple example of knowledge tracing.

To solve the problem of KT, various methods have been proposed by researchers, including the Bayesian knowledge tracing model[9, 53], factor analysis model[6, 30, 45, 48] and deep knowledge tracing model[1, 2, 26, 34]. The deep knowledge tracing model uses deep learning technology to achieve a substantial performance

improvement compared to other methods[3], so, in this paper, we mainly focus on the deep KT model. Deep KT models usually consider skills as an equivalent representation of questions and replace questions with skills as the input of the model, leading to poor model performance. However, simply changing the model input from skills to questions will lead to catastrophic failure due to the sparse interaction between students and questions[46]. Therefore, it is crucial to learn a good question representation.

Recently, contrastive learning has demonstrated powerful representation learning capabilities in the fields of computer vision and graph neural networks, and has been introduced into the field of knowledge tracing (KT) to improve question representation. Contrastive learning-based KT models[10, 35, 49] generally construct a relationship graph containing questions and use contrastive learning on the graph to obtain high-quality question representations, which significantly improves model performance. However, these methods have some limitations in selecting positive and negative samples. For example, Bi-CLKT[35] and S^2-HNN[49] use graph augmentation methods (such as removing edges or masking node features) to select positive samples, which may disrupt the semantic structure of the graph, leading to redundant and unreliable internal connections, thereby affecting the performance of contrastive learning. CDKT[10] relies on skill tags to select positive samples. However, these skill tags are usually annotated by experts and may be subject to errors or omissions. Additionally, these methods select negative samples heuristically based on the model structure. Since the performance of contrastive learning depends on the selection of positive and negative samples, selecting them in KT is a major challenge.

We attempted to introduce the BYOL[13] framework on a relationship graph containing questions. The framework eliminates the need for negative samples in contrastive learning by predicting representations of the online and target views from each other. However, traditional graph augmentation methods that obtain two different views of the original graph (the online and target views) may disrupt the semantic structure of the graph, resulting in redundant and unreliable internal connections that ultimately affect the performance of contrastive learning[40]. Instead, we propose using the original graph without modification as the online view and generating the target view through adversarial training of the node features of the online view. This approach preserves the structural information of the original graph, dynamically creates a stronger target view for the online view to promote contrastive learning, and enhances the model's robustness and generalization ability.

Specifically, we propose a novel approach for knowledge tracing called the adversarial bootstrapped question representation (ABQR) model. ABQR utilizes contrastive and adversarial learning techniques to extract rich information from questions and improve the model's robustness. We begin by converting the skill-question bipartite graph into a skill-informed question interaction graph and use it as the online view. We then use a multi-objective multi-round feature adversarial graph augmentation method to generate the target view. With this method, an optimal adversarial perturbation of the online question representation is generated through multiple rounds of joint adversarial against knowledge tracing and contrastive learning, without altering the structure of the question interaction graph. We combine the online question representation and its optimal adversarial perturbation to create the target question representation. Next, we encode the online and target question representations using the online and target graph encoders, respectively. We then allow the online and target question representations to predict each other, resulting in a high-quality question representation suitable for knowledge tracing. Our experiments on four public datasets demonstrate the effectiveness of ABQR.

Overall, our contributions can be summarized as follows:

- We introduce the BYOL framework to address the challenging problem of selecting negative samples in knowledge tracing and propose a multi-objective multi-round feature adversarial graph enhancement method to obtain a high-quality target view, which not only improves contrastive learning but also enhances the robustness of the KT model.
- We propose an adversarial bootstrapped question representation model that effectively captures meaningful question representations in the skill-informed question interaction graph.
- We conduct extensive experiments on four publicly available datasets, demonstrating the effectiveness of ABQR and its ability to outperform state-of-the-art models.

2 RELATED WORK

2.1 Knowledge Tracing

Knowledge Tracing (KT) technology was first applied in the field of education in 1995 when [9] proposed the Bayesian knowledge tracing model (BKT). BKT represents the learner's knowledge status as a set of binary variables indicating whether the learner has mastered a basic concept and updates these variables through a hidden Markov model. With the rise of deep neural networks, KT models based on deep learning gradually emerged. For instance, DKT[31] applied deep learning to KT for the first time and exhibited superior performance. DKVMN[54] used static matrices as latent basic concepts and dynamic value matrices to represent students' corresponding knowledge states. SAKT[27] introduced the self-attention mechanism into the field of KT for the first time, addressing the problem of data sparseness by identifying the correlation between concepts. SAINT[8] applied a transformer[43] to KT, making it the first encoder-decoder-based KT model. More recently, with the development of graph neural networks, researchers have proposed GNN-based KT models. GKT[25] transformed the KT task into a time-series node-level classification problem in a graph neural network, propagating the knowledge state according to the graph structure. GIKT[51] used GCN[19] to obtain the question representation on the skill-question bipartite graph for the first time. CL4KT[21] used contrastive learning[7, 16] in KT to perform data augmentation on a student's sequence of historical interactions. Bi-CLKT[35] uses contrastive learning on graphs to learn high-quality question representations to improve model performance.

2.2 Question Representation

Question representation in Knowledge Tracing (KT) has been explored through various methods. These methods can be categorized

into three main categories: (1) text-based methods, (2) methods utilizing the relationship between questions and skills, and (3) graph representation learning methods.

In the first category, text-based methods aim to capture the textual information of questions. For example, the EERNN method[38] utilizes word2vec[24] to obtain the text representation of the question based on its sequence, and bidirectional LSTM[17] is employed to extract semantic features. EKT[22] extends the EERNN approach by proposing an interpretable framework for tracing students' knowledge states across multiple concepts.

The second category focuses on the relationship between questions and skills. DHKT[46] mines the relationship between questions and utilizes it as a training constraint. qDKT[37] employs FastText[4] to derive question representations based on the sequential order of the questions. AKT[11] combines skills and questions using the Rasch model[32] to generate educational question representations. PEBG[23] leverages side information of skills and questions to learn question representations while considering question difficulty.

The third category explores graph representation learning for question representation. DKTS[47] employs graph embedding algorithms such as LINE[41] to obtain question representations. GIKT[51] applies graph convolutional networks (GCN) on the skill-question bipartite graph to acquire question representations. CoKT[39] performs random walks on the skill-question and student-question bipartite graphs and utilizes node2vec[14] to obtain question representations. Additionally, contrastive learning-based graph representation learning methods like CL4KT[21] perform data augmentation on students' historical interaction sequences. CDKT[10] considers questions containing the same skills as positive samples and questions without the same skills as negative samples. Bi-CLKT[35] employs contrastive learning on skill-skill and question-question graphs to obtain question representations.

In contrast to previous knowledge tracing models based on contrastive learning, our proposed ABQR model addresses the challenge of selecting negative samples by enabling question representations from different perspectives to mutually predict each other. Furthermore, we introduce a novel multi-objective multi-round feature adversarial graph augmentation method that generates higher-quality target views to enhance contrastive learning while preserving the integrity of the original graph structure. Through joint adversarial training for knowledge tracing and contrastive learning, our approach enhances the model's robustness and generalization capability.

3 APPROACH

In this section, we provide a detailed introduction of ABQR. Fig. 2 illustrates the overall architecture of the framework.

3.1 Problem Formulation

The KT task can be formulated as follows: Given a student's historical answer interaction $IX = \{(q_1, a_1), (q_2, a_2), ..., (q_t, a_t)\}$, where q_t represents the question answered by the student at time t, $a_t \in \{0, 1\}$ represents the student's answer, $a_t = 0$ means the student's answer is incorrect; otherwise, the answer is correct. The goal of KT is to predict the probability $P(a_{t+1} = 1|IX, q_{t+1})$ that students correctly answer the next question q_{t+1}.

3.2 Graph Transformation Module

In this section, we describe how ABQR obtains the final desired skill-informed question interaction graph.

3.2.1 Skill-Question Bipartite Graph. In the KT task, we consider two related entities: skills and questions. Each skill s_i is associated with multiple questions $\{q_1, q_2, ..., q_{m_i}\}$, similarly, each question q_i is associated with multiple skills $\{s_1, s_2, ..., s_{n_i}\}$, where m_i is denoted as the total number of questions associated with skill s_i, and n_i is denoted as the total number of skills associated with question q_i. Furthermore, we define the skill-question bipartite graph $G_{sq} = \{(s, q, r_{sq})|s \in S, q \in Q\}$, where S represents all skill sets and Q represents all question sets. When skills and questions are linked, $r_{sq} = 1$, otherwise $r_{sq} = 0$.

3.2.2 Skill-Informed Question Interaction Graph. Since our aim is to obtain question representations, for convenience, we further improve on the skill-question bipartite graph to obtain the skill-informed question interaction graph $G = \{(q_i, q_j, r_{q_iq_j}|q_i, q_j \in Q\}$, where Q represents all the question sets, when $r_{sq_i} = 1$ and $r_{sq_j} = 1$, $r_{q_iq_j}$=1, where $s \in S$, S represents all skill sets, otherwise $r_{q_iq_j} = 0$.

3.3 Adversarial Contrastive Module

We define the question feature matrix $F \in R^{N\times d}$, where N represents the total number of questions, and d represents the feature dimension. For a skill-informed question interaction graph G, we define the adjacency matrix $A \in R^{N\times N}$ of G, then graph $G = (F, A)$. The adversarial contrastive module maintains two graph encoders: the online graph encoder f_θ and the target graph encoder f_ϵ.

For the online graph encoder f_θ, we use the graph $G = (F, A)$ without modification, and then obtain the online representation $\overline{F}$ through f_θ; in other words:

$$\overline{F} = f_\theta(F, A) \tag{1}$$

Instead of computing the loss between two views using traditional contrastive methods, we predict target representations via online representations such as BYOL. We obtain the prediction $\overline{Z}$ of the target representation via the online representation $\overline{F}$, that is,

$$\overline{Z} = p_\theta(\overline{F}) \tag{2}$$

Please note that this method of predicting between samples avoids the need for negative samples.

For the target graph encoder f_ϵ, we avoid using traditional graph augmentation methods[52] to obtain another view (target view). Instead, we perturb the feature F to generate another view through the multi-objective multi-round feature adversarial graph augmentation method. Unlike ATKT[15], which only seeks adversarial perturbation once[12], we iteratively search for the optimal adversarial perturbation to maximize the model's generalization ability. The adversarial perturbation is not only against contrastive learning but also for KT. Specifically, if the contrastive learning loss is Γ_1 and the KT loss is Γ_2, then the adversarial disturbance δ is calculated as:

$$\delta = \arg \max_{\|\delta\|_\infty \le \xi} (\alpha * \Gamma_1 + \beta * \Gamma_2) \tag{3}$$

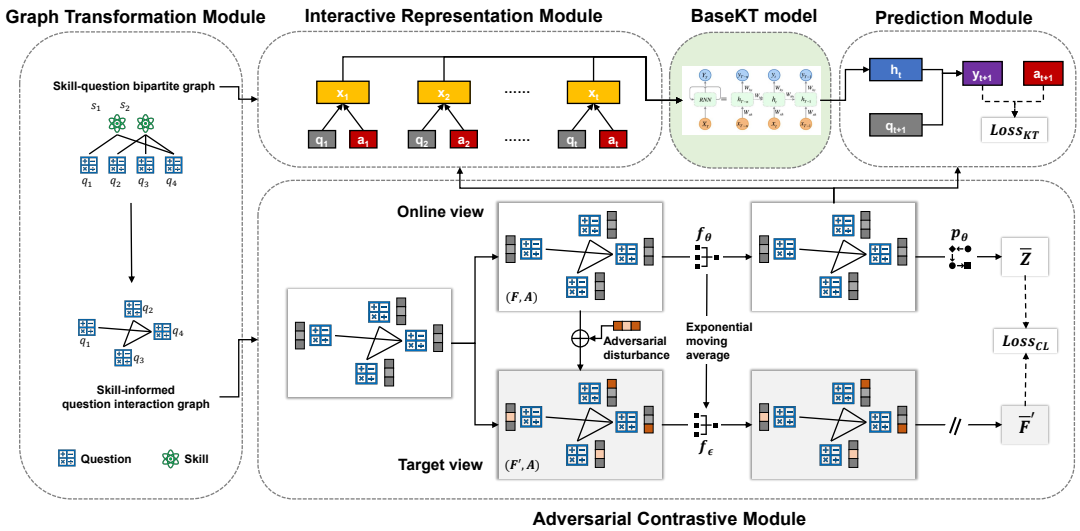

Figure 2: The overall framework of ABQR. ABQR has four main modules: graph transformation, adversarial contrastive, interaction representation, and prediction. The graph transformation module converts the skill-question bipartite graph into the skill-informed question interaction graph. The adversarial contrastive module generates target views for contrastive learning using multi-objective multi-round feature adversarial methods. The graph encoder encodes the original and target views for contrastive learning to obtain the desired question representation. The interaction representation module converts student interactions into model inputs using the learned question representations. Finally, the prediction module predicts whether the student can answer the next question correctly based on their current learning situation.

where ξ is a hyperparameter controlling the perturbation norm and α and β are hyperparameters adjusting the balance of contrastive learning loss and knowledge tracing loss, respectively. We fight adversarial training M times to find the optimal disturbance. For the g-th disturbance δ_g, we add the question feature F and the disturbance to obtain the feature representation F' of the target view, which can be formalized as:

$$F' = F + \delta_g \tag{4}$$

Then, we input F' to the target graph encoder f_ϵ to obtain the target question representation $\overline{F}'$, which can be formulated as follows:

$$\overline{F}' = f_\epsilon(F', A) \tag{5}$$

Finally, to make the prediction $\overline{Z}$ of the target question representation closer to the true target question representation $\overline{F}'$, we use the mean squared error as the loss function, specifically:

$$Loss_{f_\theta f_\epsilon} = 2 - 2 * \frac{\overline{Z}^T \cdot \overline{F}'}{\|\overline{Z}\|_2 \|\overline{F}'\|_2} \tag{6}$$

To enable two views to predict each other, we feed F and F' to the target graph encoder f_ϵ and the online graph encoder f_θ, respectively, to obtain the symmetric loss $Loss_{f_\epsilon f_\theta}$ of $Loss_{f_\theta f_\epsilon}$. Finally, the loss of contrastive learning $Loss_{cl}$ is calculated as:

$$Loss_{cl} = Loss_{f_\theta f_\epsilon} + Loss_{f_\epsilon f_\theta} \tag{7}$$

3.4 Interactive Representation Module

To obtain a more robust question representation, we use the online question representation after adversarial perturbation as the final question representation W, that is,

$$W = f_\theta(F', A) \tag{8}$$

We also initialize the answer representation matrix $K \in R^{2\times d}$, and the two rows of K denote incorrect and correct answers, respectively. W_i and K_i represent the i-th row of W and K, respectively. At the t-th moment when the student answers the question, we combine the final question representation W and answer representation K to obtain the final interaction representation X_t, specifically:

$$X_t = W_{q_t} + K_{a_t} \tag{9}$$

3.5 Prediction Module

ABQR can be easily extended to other models. Specifically, we denote the base model of KT as *BaseKT*. For a given interaction representation $(X_1, X_2, ..., X_t)$, we input it to *BaseKT* to obtain the current hidden state h_t of the student, that is,

$$h_t = BaseKT(X_1, X_2, ..., X_t) \tag{10}$$

We combine the student's current hidden state h_t and the representation of the next question $W_{q_{t+1}}$ to predict the probability y_{t+1} of the student correctly answering the next question, calculated as:

$$u_{t+1} = ReLU(W_1[h_t \oplus W_{q_{t+1}}] + b_1) \tag{11}$$

$$y_{t+1} = \sigma(W_2 u_{t+1} + b_2) \tag{12}$$

Among them, u_{t+1} is an intermediate variable, $\oplus$ is a connection operation, W_1,b_1,W_2 and b_2 are all learnable parameters, and σ is the sigmoid function.

3.6 Model Training

3.6.1 Loss Function Definition. The loss of knowledge tracing $Loss_{kt}$ is defined as the binary cross entropy loss of the predicted probability y_{t+1} and the true answer label a_{t+1}, calculated as:

$$Loss_{kt} = -\sum_{i=0}^{T-1}(a_{t+1}logy_{t+1} + (1 - a_{t+1})log(1 - y_{t+1})) \quad (13)$$

Finally, we jointly optimize the knowledge tracing loss and the contrastive learning loss. For the convenience of calculation, the total loss of the model $Loss_{total}$ is calculated as:

$$Loss_{total} = Loss_{kt} + \varphi Loss_{cl} \quad (14)$$

where φ is a coefficient that adjusts the balance between knowledge tracing loss and contrastive learning loss.

3.6.2 Parameter Updating. Adversarial perturbation multiple times jointly adversarial against knowledge tracing and contrastive learning. Specifically, the update calculation of adversarial perturbation δ is:

$$\delta_{g+1} = \prod_{\|\delta_g\|_\infty \leq \xi} \delta_g + \lambda * sign(\nabla_{\delta_g}(\alpha * \Gamma_1 + \beta * \Gamma_2)) \quad (15)$$

Among them, the adversarial disturbance δ is updated iteratively, ξ is the hyperparameter controlling the norm of the disturbance, λ is the step size of the disturbance, and α and β are the hyperparameters to adjust the balance between the knowledge tracing loss and the contrastive learning loss, respectively. This is different from Eq. 14. The calculation of adversarial disturbance comes from knowledge tracing and contrastive learning. Eq. 14 is formed for the convenience of calculation.

We iteratively update M times to obtain the optimal perturbation. However, since running M forward and backpropagations makes model training slow, we use free training[33], which parallelizes model parameter updates with perturbation updates at little additional time cost. However, the perturbation obtained by free training is suboptimal. To address this problem, we accumulate gradients[20] and apply them to update model parameters in one step.

3.6.3 Target Graph Encoder Update. To prevent contrastive learning from collapsing into a trivial solution, we update the target graph encoder f_ϵ with the exponential moving average (EMA) of the online graph encoder f_θ; in other words:

$$\epsilon = \tau\epsilon + (1 - \tau)\theta \quad (16)$$

where ϵ is the model parameter of the target graph encoder f_ϵ, θ is the model parameter of the online graph encoder f_θ, and τ is the decay rate controlling how close ϵ is to θ.

To better illustrate our model, we summarize the entire training process of the model in Algorithm 1.

4 EXPERIMENT

In this section, we present extensive experiments to answer the following questions:

- RQ1: How does ABQR compare to the state-of-the-art KT model in predicting student responses?
- RQ2: How do different components in ABQR affect its performance?

Algorithm 1 ABQR Training

Require: Graph G, student interaction IX, step size of disturbance λ and adversarial num M.

```
Initialize ABQR parameters φ and target graph encoder parameters ϵ, respectively.
while not converage do
    δ_0 ← U(−λ, λ).
    for g = 1 to M do
        Calculate Loss_cl using Eq. 7;
        Calculate Loss_kt using Eq. 13;
        Calculate Loss_total using Eq. 14;
        update δ_g using Eq. 15;
    end for
    Update model parameters φ using accumulated gradients;
    Update ϵ using Eq. 16;
end while
return φ and ϵ
```

Table 1: Summary Statistics of Processed Datasets.

	ASSIST09	ASSIST17	EdNet	Statics2011
# Student (u)	4,160	1,706	5,002	331
# Question (q)	15,680	1,150	11,775	633
# Skill (s)	167	86	1,837	106
# u-q	207,659	153,324	1,156,254	91,133
# q-s	189,370	1,150	26,345	733

- RQ3: Is ABQR sensitive to the selection of hyperparameters and GNN?
- RQ4: Has ABQR learned a meaningful question representation?

4.1 Experimental Setting

4.1.1 Datasets. To evaluate our proposed model, we conducted experiments on four datasets commonly used in knowledge tracing: ASSIST09[1], ASSIST17[2], EdNet[3], Statics2011[4]. The detailed statistics of the dataset are shown in Table 1. We do this on the dataset: for models that use skills as input, we combined multiple skills into a new joint skill[50], and for each dataset, we used 80% of all sequences as the training set and 20% as the test set.

4.1.2 Baseline Model. To evaluate the effectiveness of ABQR and show the improvement of ABQR on other knowledge tracing models, we extended DKT, DKVMN and SAKT, and we compared them with the following 8 state-of-the-art methods.

- **DKT**[31] is the first KT model based on deep learning, which traces students' knowledge status through recurrent neural networks[17].
- **DKVMN**[54] uses a storage network to store the knowledge state of students.

[1]https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data
[2]https://sites.google.com/view/assistmentsdatamining/dataset
[3]https://github.com/riiid/ednet
[4]https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507

Table 2: Performance Comparison. * indicates t-test p-value < 0.05 compared to the second best result.

Model	ASSIST09		ASSIST17		EdNet		Statics2011	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
DKT[31]	0.7684	0.7297	0.7188	0.6673	0.7006	0.7129	0.8532	0.8289
DKVMN[54]	0.7629	0.7266	0.7142	0.6639	0.6975	0.7120	0.8363	0.8224
SAKT[27]	0.7564	0.7192	0.7079	0.6596	0.6956	0.7115	0.8092	0.8111
AKT[11]	0.7888	0.7427	0.7620	0.6937	0.7647	0.7385	0.8718	0.8376
PEBG+DKT[23]	0.7738	0.7329	0.7619	0.6949	0.7571	0.7366	0.8655	0.8368
GIKT[51]	0.7859	0.7371	0.7723	0.6989	0.7640	0.7366	0.8907	0.8503
CL4KT[21]	0.7626	0.7275	0.7139	0.6615	0.6965	0.7118	0.8251	0.8170
CDKT[10]	0.7733	0.7297	0.7709	0.7019	0.7645	0.7386	0.8872	0.8510
ABQR+SAKT	0.7752	0.7312	0.7624	0.6959	0.7651	0.7382	0.8808	0.8463
ABQR+DKVMN	0.7815	0.7369	0.7693	0.7001	0.7646	0.7389	0.8941	0.8550
ABQR+DKT	**0.7906***	**0.7429**	**0.7763***	**0.7063***	**0.7683***	**0.7410***	**0.8995***	**0.8574***

- **SAKT**[27] uses a self-attention mechanism to measure the importance of previously answered questions in the current question.
- **AKT**[11] combines the attention mechanism with the Rasch model[32] to generate question representation while considering students' forgetting behavior.
- **PEBG+DKT**[23] improves the DKT model through pre-training question representation.
- **GIKT**[51] uses GCN to aggregate the relationship between questions and skills to obtain question representations.
- **CL4KT**[21] is a contrastive learning model that alleviates sparse interactions between students and questions.
- **CDKT**[10] is a contrastive learning model that considers the relationship between questions and questions.

Consistent with previous work[28, 36, 49, 55], we used the area under the curve (AUC) and accuracy (ACC) as the evaluation metrics of the model. The larger the AUC and ACC values are, the better the model performance.

4.1.3 Implementation Details. Our model was implemented using Pytorch[29]. The model was learned by the Adam[18] algorithm, the learning rate was set to 0.002, and the batch size was 80. In addition, we added the L2 norm to the weights and set the weight decay to 1e-5. To avoid model overfitting, we kept the dropout probability at 0.4. We also consistently set the gradient clipping threshold to 15.0 to avoid exploding gradients. To account for input sequences having varying lengths, we standardize them by setting a fixed length of 200. Sequences with a length less than 3 are removed since they lack meaningful information. If a sequence has a length less than 200, we fill it with zeros until it matches the fixed length. On the other hand, if a sequence has a length greater than 200, we split it into shorter sequences of the same length. The feature dimension was set to 128, and we selected other hyperparameters using grid search (see Section 4.5.1 for more details). During the experiments, it was observed that ABQR benefits from a larger ξ, and as a result, we did not impose any restrictions on ξ.

4.2 Performance (RQ1)

We implemented 5 repetitions of the experiments and report the average performance of the ABQR model in Table 2, where we highlight the best result in bold and the second best result in underline. Based on the experimental results, we made the following conclusions: First, compared to models that use only skill information as input, such as DKT, DKVMN, and SAKT, models that use question representations such as AKT, GIKT, and PEBG+DKT showed significant performance improvements. Second, there was a large performance gap between the contrastive learning model CL4KT and CDKT, which may be because CL4KT ignores the richer semantic information contained in questions. Third, after using the ABQR extension, the AUCs of the DKT, DKVMN, and SAKT models improved by an average of 4.84%, 4.97%, and 5.36%, respectively, compared to the original models. Finally, ABQR+DKT significantly outperformed the other models on all datasets, demonstrating the effectiveness of our proposed ABQR method.

Table 3: Performance comparison of ABQR and its adversarial training variant.

Model	ASSIST09	ASSIST17	EdNet	Statics2011
DKT	0.7684	0.7188	0.7006	0.8532
OAKT+DKT	0.7877	0.7755	0.7665	0.8979
OACL+DKT	0.7855	0.7721	0.7640	0.8940
ABQR+DKT	**0.7906**	**0.7763**	**0.7683**	**0.8995**
DKVMN	0.7629	0.7142	0.6975	0.8363
OAKT+DKVMN	0.7715	0.7685	0.7637	0.8925
OACL+DKVMN	0.7767	0.7629	0.7620	0.8842
ABQR+DKVMN	**0.7815**	**0.7693**	**0.7646**	**0.8941**
SAKT	0.7564	0.7079	0.6956	0.8092
OAKT+SAKT	0.7730	0.7618	0.7647	0.8778
OACL+SAKT	0.7684	0.7545	0.7640	0.8681
ABQR+SAKT	**0.7752**	**0.7624**	**0.7651**	**0.8808**

4.3 Adversarial Training Analysis (RQ2)

To evaluate the effectiveness of our proposed multi-objective adversarial training method, we conducted experiments comparing ABQR with two variants:

- **OAKT** Only Adversarial against Knowledge Tracing.
- **OACL** Only Adversarial against Contrastive Learning.

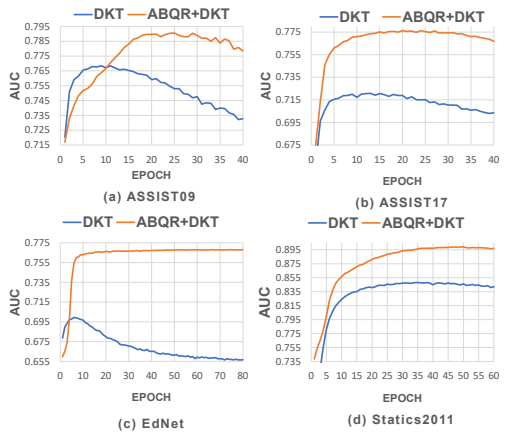

Figure 3: Convergence curves of DKT and ABQR+DKT.

We evaluated the models using AUC as the performance metric, and the results are shown in Table 3.

Table 3 reveals that the performance of OAKT is slightly better than that of OACL, indicating that adversarial training is helpful for the knowledge tracing task. Although OACL is beneficial for contrastive learning, its assistance in knowledge tracing is limited. On the other hand, ABQR, which combines both adversarial against knowledge tracing and contrastive learning, achieved the best performance, demonstrating the effectiveness of the multi-objective adversarial training method. Furthermore, the convergence curves of DKT and ABQR+DKT on the four datasets in Fig. 3 show that ABQR+DKT can effectively improve the robustness of DKT and significantly improve the performance of DKT.

Table 4: Performance comparison of ablation study.

Model	ASSIST09	ASSIST17	EdNet	Statics2011
DKT	0.7684	0.7188	0.7006	0.8532
w/o MMFA+DKT	0.7836	0.7707	0.7615	0.8944
w/o CL+DKT	0.7796	0.7750	0.7660	0.8972
ABQR+DKT	**0.7906**	**0.7763**	**0.7683**	**0.8995**
DKVMN	0.7629	0.7142	0.6975	0.8363
w/o MMFA+DKVMN	0.7727	0.7611	0.7615	0.8831
w/o CL+DKVMN	0.7727	0.7686	0.7636	0.8939
ABQR+DKVMN	**0.7815**	**0.7693**	**0.7646**	**0.8941**
SAKT	0.7564	0.7079	0.6956	0.8092
w/o MMFA+SAKT	0.7654	0.7564	0.7635	0.8706
w/o CL+SAKT	0.7677	0.7609	0.7649	0.8780
ABQR+SAKT	**0.7752**	**0.7624**	**0.7651**	**0.8808**

4.4 Ablation Studies (RQ2)

In this section, we conduct an ablation study to explore the impact of each module in ABQR on the model's performance. We compared ABQR with its two variants, which are described below. We used AUC as the evaluation metric, and the performance is shown in Table 4.

- **w/o MMFA**, which removes the Multi-objective Multi-round Feature Adversarial graph augmentation method. At this time, ABQR uses traditional graph enhancement methods (removing edges and masking node features) to generate another view.
- **w/o CL**, which removes the Contrastive Learning loss of the model but still retains the adversarial contrastive module.

Based on the results presented in Table 4, we found that w/o MMFA resulted in the worst performance, indicating the crucial role of the multi-objective multi-round feature adversarial graph augmentation method in enhancing the generalization ability of the model. Similarly, w/o CL resulted in similar performance degradation, indicating the importance of good question representations in enhancing the model's performance. On the other hand, ABQR+DKT, ABQR+DKVMN, and ABQR+SAKT, which incorporate all the modules, achieve the best performance, indicating the effectiveness of each individual module.

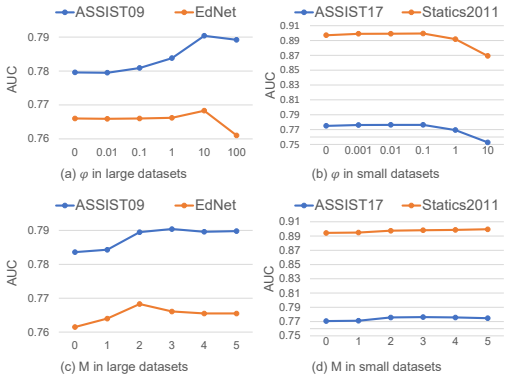

Figure 4: Effect of hyperparameters on ABQR.

4.5 Sensitivity Analysis (RQ3)

4.5.1 Hyperparameter Analysis. In this section, we explore the impact of two hyperparameters on ABQR: the balance coefficient φ and multi-round adversarial number M. We conducted experiments using ABQR+DKT on four datasets and used AUC as the evaluation metric. The results are plotted in Fig. 4.

First, regarding the balance coefficient φ, setting φ to 0 means not using contrastive learning. As shown in Fig. 4(a), when the dataset is large, such as ASSIST09 and EdNet, a larger balance coefficient leads to better performance. The optimal value of φ is 10. However, when the dataset is small, a smaller balance coefficient leads to better performance, as shown in Fig. 4(b). It is evident that obtaining high-quality question representations through contrastive learning is highly beneficial for the model.

Second, regarding the multi-round adversarial number M, setting M to 0 means adversarial training is not used. As shown in Fig. 4(c) and Fig. 4(d), when the number of adversarial trainings was less than or equal to 3 on both large and small datasets, the performance of the model significantly improved as the number of adversarial trainings increased. As the number of times increased, the model's robustness became sufficient, and further increasing the number of adversarial trainings may not significantly improve the model's performance.

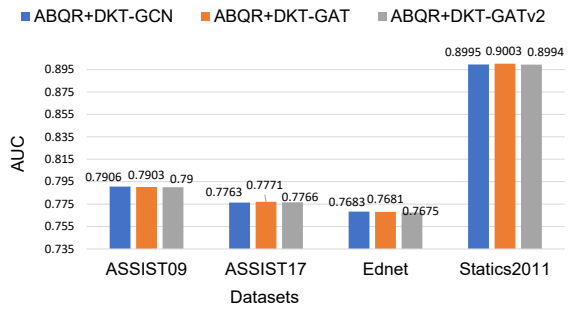

Figure 5: Impact of GNN on ABQR.

4.5.2 GNN Analysis. In this section, we investigate the sensitivity of ABQR to GNNs. We conducted experiments using ABQR+DKT on four datasets and compared three GNN encoders: GCN[19], GAT[44], and GATv2[5]. We evaluated the performance using AUC as the metric and present the performance statistics in Fig. 5.

Based on the results shown in Fig. 5, we found that for the ASSIST09 and Ednet datasets, GCN performed slightly better than the other models. For the ASSIST17 and Statics2011 datasets, GAT performed slightly better than the others. GATv2, on the other hand, exhibited moderate performance across all datasets. However, the performance differences between the three models were not significant, indicating that ABQR is not highly sensitive to the choice of GNN encoder. This is likely because ABQR is not sensitive to the skill-informed question interaction graph, which also suggests that ABQR may still be applicable to other question-based graphs. Since the GAT and GATv2 models consumed more memory and took longer to run, the default GNN encoder used in this experiment was GCN, unless otherwise specified.

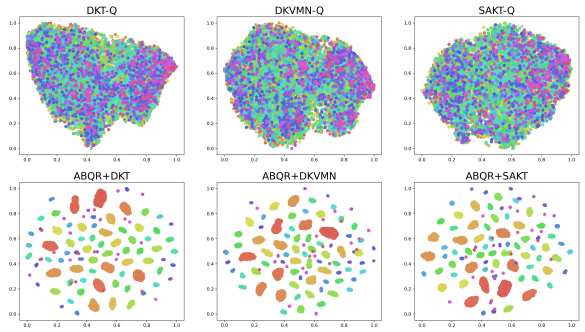

Figure 6: Visualization of question representations. Questions with the same skill are marked with the same color.

4.6 Visualization (RQ4)

In this section, we show that ABQR improves upon the base KT model and learns high-quality meaningful question representations. To obtain the question representations of the base KT model, we replaced the original skills with questions as inputs to DKT, DKVMN, and SAKT, which are recorded as DKT-Q, DKVMN-Q, and SAKT-Q, respectively. We used t-SNE[42] to project the resulting question representations into a 2D space and visualize them on the ASSIST09 dataset. Questions with the same skill are marked with the same color. Fig. 6 shows the visualization results.

We found that the base KT model could not learn the relationship between skills and questions. The questions learned by the base KT model were scattered randomly. In contrast, ABQR learns the question representation, where questions containing the same skills are closely grouped together, and questions without the same skills are effectively separated. This indicates that ABQR learned more meaningful question representations than the base KT model.

5 CONCLUSION

In this paper, we proposed ABQR, an adversarial bootstrapped question representation model for knowledge tracing. First, we constructed a skill-question interaction graph from the skill-question bipartite graph and learned question representations from different views of the interaction graph. These representations were trained to predict each other, eliminating the need for negative sample selection. We also proposed a multi-objective multi-round feature adversarial graph augmentation method that generates a high-quality target view to facilitate contrastive learning while preserving the original graph structure. This adversarial method against knowledge tracing and contrastive learning many times greatly improves the robustness of the model. Our experiments demonstrated that ABQR outperformed state-of-the-art models on all datasets.

6 LIMITATIONS

During our experiments, we discovered that ABQR can be effectively integrated into existing knowledge tracing models as a plugin, leading to enhancements in their predictive performance. However, the degree of improvement varies depending on the specific knowledge tracing model used as the base model. ABQR, for example, achieves substantial performance improvement for DKT. We believe that there is a certain correlation between the question representation method and the base model structure. For different knowledge tracing base models, the optimal question representation method or representation model parameter combination may be different.

In future work, we intend to investigate the correlation between the base models with different structures and the sensitivity of question representation learning. We believe that this will facilitate the development of improved knowledge tracing base models and methods for question representation learning.

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